

How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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CS-503 Visual Intelligence — Project Proposal

Abstract—Cognitive maps emerge in the recurrent hidden states of navigation agents, as shown for both sighted and blind conditions [1], but all prior settings keep visual certainty spatially constant, bypassing the need to manage perceptual uncertainty. We introduce a foveated sensor—sharp at the centre, degraded in the periphery—and probe whether the resulting spatial memory merely buffers observations or develops compensatory representations of the agent’s own observational uncertainty. We further ask whether foveation drives a shift toward allocentric spatial codes by forcing integration across partial glimpses, and whether learned gaze targets regions where spatial memory is weakest. We train five agents with an identical LSTM backbone on PointGoal navigation in Habitat, varying only visual input (blind, uniform, foveated-fixed, foveated-learned, matched-compute). To investigate the resulting memory structures, we decode spatial variables from frozen hidden states via linear probes, compare representational geometry across conditions with CKA, and identify place-cell-like units through per-unit rate maps.

I. INTRODUCTION

Wijmans et al. [1] showed that RL agents trained on PointGoal navigation develop map-like representations in their recurrent hidden states—even *blind* agents receiving only goal displacement, GPS, compass, and a close-to-goal scalar. Dwivedi et al. [2] probed sighted agents for related spatial concepts. However, all these settings keep visual certainty spatially constant, leaving open whether the spatial representations are a property of the navigation task or an artifact of uniform input quality.

In biological vision, input quality is never uniform: humans see sharply only within $\sim 2^\circ$ of gaze. Wirth et al. [3] showed that primate hippocampal spatial representations depend on what the animal foveates, not just where it is—suggesting that spatial memory is not merely shaped by where an agent goes, but by how reliably it observes each region. If this holds for learned agents, a foveated sensor should not merely degrade cognitive maps but reshape them: memory may develop qualitatively different strategies to compensate for uneven observation quality.

A. Problem Statement

These two accounts—passive buffering vs. active compensation—make divergent predictions. A passive memory’s accuracy should fall with eccentricity, since peripherally-seen regions are stored less faithfully. An uncertainty-aware memory predicts the opposite: accuracy staying flat or *rising*, because the agent devotes representational capacity to uncertain regions. We ask: **does a**

foveated agent’s memory follow the passive-drop pattern, or does it exhibit the compensatory signature of epistemic representation?

B. Motivation

- **Epistemic memory from task pressure alone.** If foveation forces memory beyond passive buffering, task loss alone can induce principled uncertainty management—a concrete inductive bias for memory design in partially observable domains.
- **Allocentric mapping from restricted vision.** Foveation removes the egocentric view-matching shortcut available to uniform-vision agents, potentially making perceptual bottlenecks a sufficient condition for viewpoint-invariant spatial codes.
- **Gaze as epistemic action.** If the agent looks where its memory is weakest, this connects learned gaze to the information-gain framework of biological active vision [4], moving beyond treating gaze as a black-box performance lever [5].

II. METHOD

A. Experimental Setup

We train all agents on PointGoal navigation in Habitat [6] using a merged pool of Gibson-0+ [7] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes, following Wijmans et al. [1] Appendix A.1. Evaluation uses 18 held-out MP3D test scenes (1008 episodes), providing an out-of-dataset generalization test.

All agents share an identical 3-layer LSTM (512-d) and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. The conditions differ only in visual input, so observed differences in memory content map directly to visual input structure:

- **Blind:** no visual input (replicates [1]).
- **Uniform:** 256×256 RGB via a ResNet-18 encoder.
- **Foveated (fixed gaze):** 256×256 RGB with eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) centred at the image midpoint.
- **Foveated (learned gaze):** blur centre predicted by a lightweight MLP ($\sim 33K$ params) reading the previous LSTM state, trained end-to-end.
- **Matched-compute:** uniform 48×48 RGB (2,304 pixels, exceeding the foveated effective budget of $\approx 1,816$

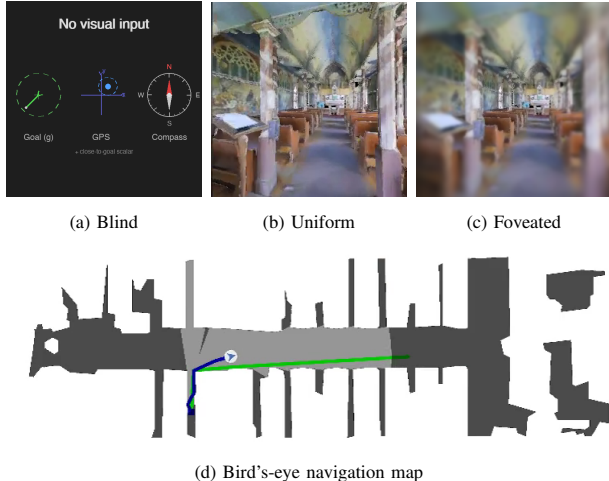


Figure 1. Matterport3D test episode. (a) Blind: no visual input; only the non-visual sensor stack (goal-in-start-frame g , GPS, compass, close-to-goal scalar). (b) Uniform: 256×256 RGB. (c) Foveated: eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff). (d) Agent trajectory (blue) vs. geodesic shortest path (green).

pixels), controlling for total information volume rather than its spatial distribution.

B. Analysis Framework

Evaluation proceeds in two stages. First, we confirm *task competence*: each trained agent is tested on the 1008 held-out MP3D episodes, measuring Success (fraction of episodes where the agent reaches the goal within a distance threshold) and SPL [8] (Success weighted by Path Length, zero for failed episodes). This ensures all five agents are competent navigators before we analyse their representations.

Second, we analyse *what the hidden states encode*. We freeze each agent’s weights, roll out episodes, and collect per-step LSTM hidden states paired with ground-truth labels (agent position, heading, GPS, observation eccentricity). We then apply multiple complementary analysis methods, guarding against known limitations of any single approach [9]:

(i) Linear probing with controls: Ridge regression from frozen hidden states to spatial targets [1], validated via control tasks [10], which assign deterministic-but-random targets per input type; a selective probe must achieve high accuracy on the real task but low accuracy on the control, ensuring the probe captures genuine structure rather than memorisation. **(ii) CKA:** Centered Kernel Alignment [11] compares representation geometry across conditions without training decoders, revealing whether foveated and uniform agents learn similar or divergent internal spaces. **(iii) Per-unit rate maps:** each LSTM unit’s activation is plotted against agent position [12], identifying place-cell-like neurons; ablating high-selectivity units tests whether they are causally involved in navigation performance (though distributed coding may limit single-unit effects). **(iv) Visualization:** PCA/t-SNE of hidden states coloured by position,

condition, and eccentricity to provide qualitative insight into representational structure.

With these tools in hand, we organize the analysis around three hypotheses, each with a concrete test using the probing infrastructure above.

H1 (Epistemic compensatory memory) Hidden states actively encode observation quality alongside layout to compensate for weak input. Tested via two methods: probing for a ground-truth “confidence” target (inverse cumulative blur σ), and stratifying layout-probe accuracy by eccentricity to show accuracy *rises* in the periphery—a pattern impossible for a passive buffer. The null prediction is that layout-probe accuracy simply degrades with eccentricity (tracking input quality), and that the confidence probe achieves no better accuracy than its control task—the agent tracks layout but not how well it observed each location.

H2 (Allocentric vs. Egocentric shift) Foveation forces a shift from egocentric view-matching to an objective allocentric map. Tested by evaluating the cross-heading generalization of position probes: a true allocentric map will maintain high positional decoding accuracy even when facing novel directions, an invariance we expect to be stronger in foveated agents.

H3 (Epistemic gaze coupling) Learned gaze targets high-uncertainty spatial regions to maximize information gain. Tested by correlating gaze coordinates with regions of highest recent probe error (a memory uncertainty proxy). Furthermore, if learned gaze amplifies H1–H2 effects compared to the fixed-centre ablation, active fixation is directly driven by internal epistemic demands.

C. Milestone 1: Training and baseline probing (Weeks 1–4)

Weeks 1–2 (completed): Pipeline implementation. Four-condition training (blind, uniform, foveated-fixed, matched-compute) launched on SCITAS. Probing pipeline validated with intermediate checkpoints.

Weeks 3–4: Training convergence for all four conditions. Learned-gaze foveated training launched. MP3D out-of-dataset evaluation. Per-unit rate maps and initial visualisations.

D. Milestone 2: Full analysis and deliverables (Weeks 5–8)

Weeks 5–6: Full H1–H3 probing suite across all five conditions. CKA cross-condition comparison. Control-task validation. Gaze ablation analysis.

Weeks 7–8: Occupancy-map decoding (stretch goal). Final evaluations, project webpage, and presentation.

III. RELATED WORK

Cognitive maps in RL agents Wijmans et al. [1] showed that spatial maps emerge in blind agents’ LSTM states. Banino et al. [12] and Cueva & Wei [13] independently found grid-cell-like representations in recurrent

networks trained on path integration, and Gornet & Thomson [14] extended this line using visual predictive coding. Dwivedi et al. [2] probed PointNav agents for spatial concepts (reachability, progress, target visibility) and identified which neurons encode them. These studies all use uniform or absent perception—none vary the spatial quality of visual input.

Foveated vision in deep learning Mnih et al. [15] introduced recurrent attention models, an early instance of learned gaze in deep nets. Pourrahimi & Bashivan [16] trained a foveated visual-search model and probed for neuroscience-predicted features (V1–IT); Shang & Ryoo [5] jointly learned gaze and motor policies under limited observability. These study visual search or task performance, not how a navigation agent’s recurrent memory evolves over time.

Neuroscience motivation Wirth et al. [3] showed that primate hippocampal spatial representations depend on what the animal foveates, not just where it is: foveating landmarks strongly enhances hippocampal landmark selectivity. Atanov et al. [17] showed computationally that sensor layout changes downstream task performance.

These three lines converge on a single untested prediction: spatially non-uniform input should reshape the internal representations that navigation agents learn. Our work provides the first direct test of this prediction in learned embodied agents.

IV. DISCUSSION

A. Implications and broader impact

If H1 holds, the foveated agent’s memory actively tracks its own observation quality and compensates for weak input—evidence of principled epistemic representations under partial observability [18]. If H2 holds, it confirms that restricted vision is a key driver for true allocentric mapping. If H3 holds, the learned gaze policy acts as an active information-gathering mechanism driven by internal epistemic demands. A complete null across all three hypotheses would indicate that the primate foveal–hippocampal link [3] does not transfer to learned agents under these conditions—a useful boundary for the analogy.

Mixed outcomes are equally informative. For instance, if H1 holds but H2 does not—the agent compensates for peripheral uncertainty yet still relies on egocentric representations—this would suggest that epistemic memory and allocentric mapping are dissociable, and that foveation alone is insufficient to force heading-invariant spatial codes. Conversely, if H2 holds but H1 does not, foveation may drive allocentric mapping through the need to integrate across viewpoints without the memory explicitly tracking observation quality. Either pattern constrains the design of memory modules for embodied agents under partial observability.

B. Potential risks and shortcomings

Design scope We use a simple foveation model (Gaussian blur), gaze (deterministic MLP), and the Wijmans LSTM backbone across all conditions. This limits generality but ensures any observed effect reflects spatial non-uniformity rather than architecture. Richer alternatives—log-polar transforms, stochastic gaze, Transformer memory—are well-motivated follow-ups.

Learned-gaze collapse The learned-gaze MLP (~33K parameters, <1% compute overhead) is the highest-risk component: it could converge to a fixed point, rendering the foveated-learned condition equivalent to foveated-fixed. The fixed-centre gaze ablation doubles as a fallback, so H1 and H2 remain testable even if the gaze policy fails to learn.

Compute and data Gibson and MP3D (both under signed data-use agreements) are on SCITAS. Sighted agents train for ~250M steps (~3–4 days on 2 V100 GPUs); the blind agent trains for ~500M steps (~5–7 days). Five conditions fit within the cluster allocation. All software (Habitat-sim 0.3.3, DD-PPO, PyTorch 2.1) and the probing pipeline are implemented and tested.

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